Video Forgery Detection using CNN

Litty Koshy
SCMS School of Engg. and Technology
Kochi, Kerala
littykoshy@scmsgroup.org

Ajay S
SCMS School of Engg. and Technology
Kochi, Kerala
ajaysan2017@gmail.com

Akhil Paul
SCMS School of Engg. and Technology, Kochi, Kerala
akhilthomaspaul@gmail.com

Hariharan V
SCMS School of Engg. and Technology
Kochi, Kerala
hariharanv72@gmail.com

Ashil Basheer
SCMS School of Engg. and Technology
Kochi, Kerala
ashilkb848@gmail.com

Abstract--*With the widespread use of digitally interactive multimedia such as audio, images, and video, there has been a significant increase in the mode and motivation to create digital forgeries. The widespread availability of video information and services, as well as the low cost of devices such as cameras, camcorders, and CCTVs, has led to widespread use of video information and services in our society for a variety of purposes such as video surveillance, forensics investigation, and entertainment. Previously, video editing techniques were mostly employed to improve digital information. However, as the popularity of low-cost, easy-to-use video editing software has grown, so has the number of negative repercussions and risks associated with such editing procedures. By merging, changing, or synthesising new footage, video forgery is a technique for creating changed or fraudulent videos. A method based on deep learning is given in the proposed system for classifying videos as tampered or original. The video clip that is used as input is divided into two categories: original and modified. The video is segmented into non-overlapping frames, and the authenticity of the movie is determined by whether or not all of the frames are genuine. The suggested method uses a deep CNN model that has two types of layers: (1) CNN layers which involve convolutional, pooling and fully connected layers and (2) Parasitic layers.*

Keywords:Video forgery,CNN,Parasitic,tampered

I . INTRODUCTION

In the present digital age of internet and social media, our day to day life is permeated with digital video content as one of the prominent means for communication.Developments in video technologies such as generation, transmission, storage and retrieval along with applications like Video sharing platforms, Video-conferencing etc have served the people and society in many ways. Applications such as the entertainment business, video surveillance, legal evidence, political films, video tutorials, ads, and other social networking platforms, such as YouTube, Facebook, and Instagram, demonstrate their unparalleled role in today's context.

Videos can be made, stored, transmitted and processed in digital format in an easy way, because of extensive use of the Internet and inexpensive and high quality cameras, computers and user-friendly editing tools. Any novice individual can utilize these techniques to make unauthorized modifications to the video content thereby affecting its integrity and authenticity. This possibility raises the need to validate whether the multimedia content available on the internet, obtained as a part of a video surveillance system, or received by a broadcaster, is original or not.Digital Video Forgery is the intentional modification or alteration of a digital video for the purpose of fabrication. Manipulation of a video in such a way that the content is perceptually altered is referred to as video forgeries. Spatial forgeries and Temporal forgeries are the two types of video forgeries. The field of digital video forensics was created to address the problem of counterfeiting and assure the authenticity of digital videos. Digital video forensics is a set of tools and procedures that may be used to determine whether or not the contents of a digital video are verifiable. Digital video forensics, also termed as Video Forgery Detection techniques is the scientific understanding and skill necessary to authenticate and enhance a video. Video Forgery can be performed in various ways. The different types of tampering that are applied to videos are:

i) Low Level Tampering

The picture is detected from videos and then moved to another location or manipulated in the picture in this sort of tampering. At the temporal or spatial level, this manipulation is employed.

ii) Frame by Frame Level Tampering

The frames from videos are retrieved first, and then they are tampered with. The forger may change the contents of movies by removing, adding, or copying frames. It's one of the major temporal tampering methods that can be implemented to change frames in videos.

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iii) Mass Level Tempering

The tampering is done on video blocks, which are defined as any set of video frames. In videos, these blocks are chopped and replaced. At the block level, it's called spatial tampering.

iv) Voxel Level Tempering

Video frames are modified at the pixel level in this method. Video voxels are changed, copied, or replaced. Spatial attacks are carried out at the pixel level.

The goal of video forgery detection is to expose and examine the hidden information about a film. It can be divided into two categories namely

i) Active video forgery detection

In active video forgery detection, techniques such as Digital Watermarking and Digital Signatures are utilised to validate Content Ownership and Copyright Violations.

ii) Passive video forgery detection

As opposed to the Active approach, the passive method does not necessitate the use of specialist hardware and does not necessitate any direct knowledge of the video's contents. As a result, it's also known as the Passive-Blind Approach.

The method proposed here essentially consists of the following steps

- The video is split into non-overlapping frames, and the authenticity of the clip is determined by whether or not all of the frames are genuine.
- After that, the retrieved characteristics are handed on to the CNN layers, which calculate the video's hierarchical representation.
- Finally, the custom CNN layers(parasitic), which act as a classifier, forecast whether the films are tampered or original.

The remaining sections are ordered as follows; Section II explains the literature survey, section III explains the proposed architecture, section IV explains the results and observations, and section V includes conclusion.

II. LITERATURE SURVEY

Detecting forgeries in videos has become an increasingly important concern that needs to be addressed. Some of the methods that have been employed for the detection are

Michihiro Kobayashi, Takahiro Okabe, and Yoichi Sato suggested Detecting Video Forgeries Based on Noise Characteristics in [1]. The authors of this study claimed that there are two types of approaches for producing a scene with a static background. The first is to replace frames or areas with duplication from an existing video sequence: by overwriting certain frames, forgers might conceal untrustworthy objects.The second is to clip items from other video segments and apply those to the regions which are required. Here the main objective is to show items that contribute to false evidence. Using duplication can assist in returning high correlation with original frames and duplicate frames.

In [2],proposed Correlation of Noise Residue for the detection of Video Forgery .Sensor noise pattern had effectively exploited as intrinsic sensor biometrics in recent years. The suggested technique first extracts the pattern noise images from training photos collected with any of the specific cameras, and then uses averaging operations on the pattern noise images to obtain the reference pattern noise image. The sensor pattern noise was used to identify scanner models and detect manipulation in scanned images. In addition, the sensor pattern noise was originally used to detect forgeries in photographs. To extract the sensor pattern noise of the cameras, the following strategies have to obtain a number of photos acquired from any of the existing video cameras.

In [3] proposed the detection and localisation of video copy-moves has been the subject of several investigations. The correlation coefficient is used as a measure of similarity to detect large copy-moved chunks, with the same method being expanded to include spatio-temporal blocks. However, if the cloned area is vast and has not been post-processed, this approach works effectively. Performance suffers when compression, blurring, geometric transformations, and intensity shifts occur. This is significant since these processes are usually necessary to make forgeries more realistic, and they can even be done on purpose by a professional forger to deceive forensic instruments.

In [4],proposed Forgery Detection Algorithm for Video Foreground Removal. Because even a short video might contain hundreds of frames,the challenges of algorithms in digital video forensics must address the issues of detection accuracy and processing cost.The frame-by-frame methods in previous detection algorithms resulted in inefficiency. Videos with complicated backgrounds were similarly undetectable (e.g., Unstable screens, moving trees, water waves, noise and change in brightness). The research proposes a unique forgery detection method for video foreground removal to overcome these issues. The energy factor (EF) of each frame is determined first to identify the forged frames. In the first stage, the adaptive parameter-based visual background extractor method(AVIBE) is utilised to find suspicious regions in the counterfeit frames. The programme calculates the EF difference between suspicious locations in counterfeit frames and similar regions in actual frames, and then locates the tampering evidence.

In [5], proposed Spatial Video Forgery Detection and Localization using Texture Analysis of Consecutive Frames. The detection of forged video segments (VSs) as well as the location of forged frames are the

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subject of this research. The texture of micro-patterns in tampered frames is changed during spatial domain video tampering, which is a key clue for detecting this sort of fraud. To represent the tampering traces in video frames, Subramanyam and Emmanuel used the texture descriptor Histogram of Oriented Gradients (HOG). Gradient orientation is used in this descriptor and its variants to efficiently express local texture micro-patterns and changes.

III. PROPOSED ARCHITECTURE

A technique based on deep learning is provided in the proposed system for classifying videos as tampered or original. The video clip that is used as input is divided into two categories: original and modified. The video is split into non-intersecting frames, and the authenticity of the movie is determined by whether or not all of the frames are genuine. The suggested technique uses a deep CNN model that has two types of layers: (1) CNN layers (convolutional, pooling, and dense layers) and (2) custom CNN layers(parasitic). Following frame creation, retrieved features are transferred to the CNN layers, which calculate the video's hierarchical representation.Finally, the custom CNN's dense layer, which acts as a classifier, determines whether the clips are tampered or original. The following figure depicts the entire architecture.

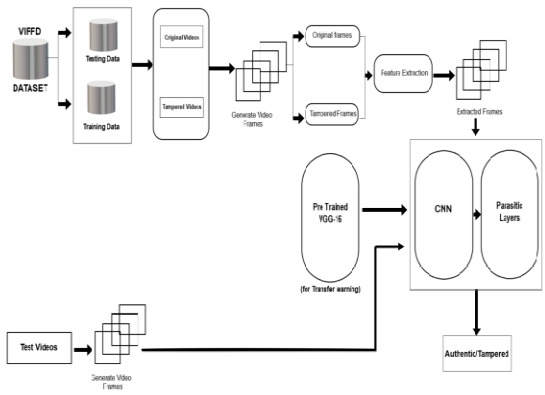

Fig 1. Proposed architecture

A)Learning Algorithm

Learning Algorithm for the proposed architecture is explained as follows.

1. The video clip that is used as input is divided into two categories: original and modified.
2.Input :The video is split into non-intersecting frames(VC) of size W.
3. Feature Extraction:
   3.1 Extract the frames
   3.2 Resize the Frames to 128*128*3.
4.Transfer learning using deep CNN Model.
5.Output:Classified as Original or tampered videos.

i) Preprocessing and Feature extraction

Since a classification architecture to train on a video classification dataset is being used, the dataset must be preprocessed first. Primarily three steps are being done in preprocessing; Extract frames, Resize and Normalize frames. In order to be able to classify the videos, the videos are first divided into non overlapping frames. OpenCV provides a very simple interface to do this. The videos are converted into frames using python OpenCV bindings. The generated frames are then resized to a specific height and width. By definition, the larger the image, the more the data, and therefore the longer it takes for algorithms to process the data. So these dimensions are adjusted, which in the proposed system is 128x128 pixels. After resizing, the resized frames are normalized by dividing with 255 so that each pixel value then lies between 0 and 1. An empty list is declared to store all the video frames and all the frames are appended to that list.

ii)Transfer learning

It is a deep learning approach in which an existing model is taken that has been trained on a large amount of data and the characteristics acquired from that data are applied to our problem. That model has been well-trained to discover some features because it has learned from a large amount of data. A portion of the trained model can be changed and adapted to our needs. Instead of training all of the layers of the model, transfer learning locks/freezes parts of them and uses the trained weights in the locked/freezed layers to extract specific characteristics from the data. All the lower layers need not be locked, some of the lower layers can be retrained because those layers will be specialised for the data. Neural networks are used in computer vision to determine the edges in the first layer, form in the intermediate layer, and task-specific features in the latter layers. In transfer learning, the initial and intermediate layers are used, and the succeeding layers are only retrained.It uses the tagged data from the task on which it was trained. In transfer learning, it can be used to transfer as much knowledge as possible from the prior task on which the model was trained to the new task at hand. This information can take many various shapes depending on the situation and the data.It could be the way models are made, for example, which makes it easier to recognise new items.Transfer learning provides a number of advantages, the most important of which are reduced training time, improved neural network performance (in most circumstances), and the absence of a large amount of data. A lot of data is normally required to train a neural network from the beginning, but it's not always possible to access data and this is when transfer learning can come into use. Transfer learning can build an effective machine learning model with relatively little training data because the model has previously been pre-trained.This is especially useful in natural language processing, where huge labelled

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datasets require a lot of expert knowledge. Additionally, training time is minimised because building a deep neural network from start on a challenging job can take hours,days or weeks.

iii)Parasitic layers

The input layer and the last three layers of the VGG-16 model (two dense, namely FC1 FC2, and the softmax) are changed in the proposed technique.The input layer of the VGG-16 model by default accepts an input image shape of 224x224.This layer was replaced by a layer accepting an input shape of 128x128.The last layers were replaced by a custom CNN layer consisting of 8 layers. The final dense layers of the VGG-16 learn task-specific information and are trained using the SGD algorithm, which requires a lot of data and a lot of time. The VGG-16's FC1, FC2, and Softmax layers have been replaced by eight new parasitic layers, which are dependent on the activation of the previous layer. These layers improve classification efficiency and accuracy by learning the video dataset's unique properties. In the proposed technique, CNN layers and parasitic layers are learnable layers. The CNN layers are pre-trained on the ImageNet dataset, and the learnable section only includes parasitic layers with a small number of parameters and enough data to perform the task.

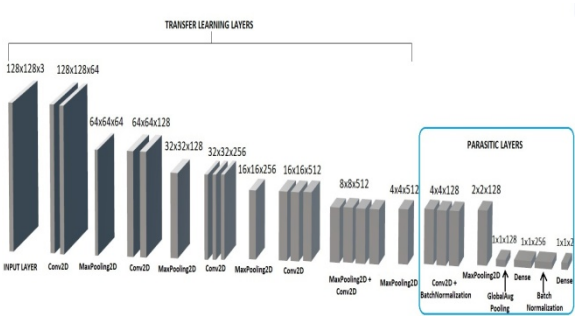

Fig 2. Custom CNN model

The additional 8 parasitic layers consist of two conv2D layers, two Batch Normalization layers, a MaxPooling2D layer, a GlobalAvgPooling layer and 2 Dense layers. Batch normalization is a transformation that keeps the mean output close to 0 and the standard deviation of the output close to 1. The pool size is still set to the size of the layer input when using Global Average Pooling, but instead of the maximum, the pool's average is used. They're frequently employed in classifiers to substitute fully-connected or densely-connected layers.Each neuron in the dense layer receives input from all neurons in the previous layer, making it a deep-connected neural network layer. The output of the dense layer is an 'm' dimensional vector. As a result, a dense layer is mostly employed to alter the vector's dimensions.Here the dense layer is associated with the classification of output.Softmax takes a set of values and generates a probability distribution. Softmax is widely used as the activation for the last layer of a classification network since the output can be interpreted as a probability distribution.

IV . RESULTS AND DISCUSSION

i)observations:

Figure 3 and 4 shows the increase in accuracy achieved by the custom model over the already existing VGG-16 model.

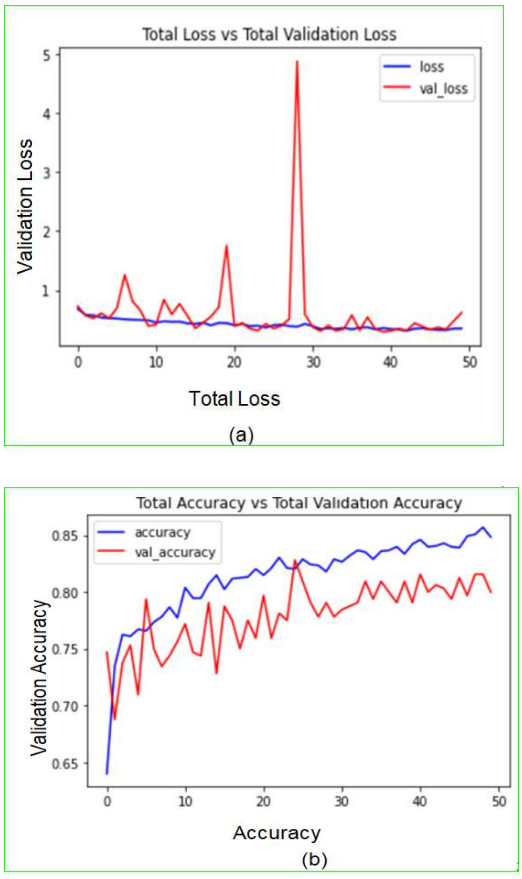

Fig 3. Plots of VIFFD trained on VGG-16 .(a) Total Loss Vs validation Loss (b) Accuracy Vs Validation Accuracy

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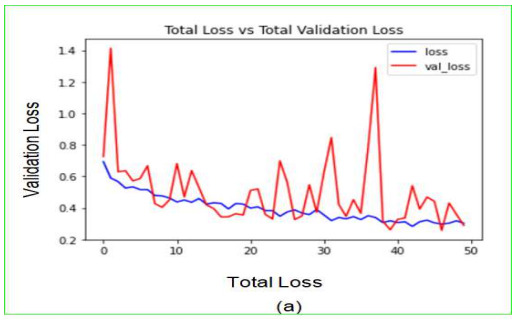

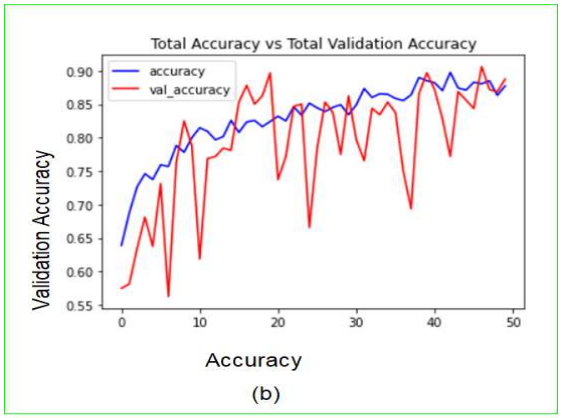

Fig 4. Plots of VIFFD trained on custom CNN model. (a) Total Loss Vs validation Loss (b) Accuracy Vs Validation Accuracy.

From the plots(Fig 3 and Fig 4) above it can be seen that VIFFD trained on the custom CNN model results in better accuracy and a much lesser loss. We have trained our custom CNN model upto 50 epochs.The Total loss vs Total Validation loss plot of VIFFD trained on VGG-16 has values ranging from 1 to 5, the same plot of VIFFD trained on custom CNN model has loss values in the range 0.2 to 1.4 which implies that the loss values of the custom CNN model are far lower than VGG-16 model.

```
Epoch 40/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3182 - accuracy: 0.8852 - val_loss: 0.2613 - val_accuracy: 0.8969
Epoch 41/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3064 - accuracy: 0.8820 - val_loss: 0.3270 - val_accuracy: 0.8719
Epoch 42/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3119 - accuracy: 0.8703 - val_loss: 0.3364 - val_accuracy: 0.8281
Epoch 43/50
320/320 [==============================] - 3s 9ms/step - loss: 0.2825 - accuracy: 0.8977 - val_loss: 0.5417 - val_accuracy: 0.7719
Epoch 44/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3118 - accuracy: 0.8742 - val_loss: 0.3920 - val_accuracy: 0.8687
Epoch 45/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3220 - accuracy: 0.8711 - val_loss: 0.4686 - val_accuracy: 0.8562
Epoch 46/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3048 - accuracy: 0.8828 - val_loss: 0.4421 - val_accuracy: 0.8438
Epoch 47/50
320/320 [==============================] - 3s 9ms/step - loss: 0.2973 - accuracy: 0.8805 - val_loss: 0.2575 - val_accuracy: 0.9062
Epoch 48/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3030 - accuracy: 0.8852 - val_loss: 0.4311 - val_accuracy: 0.8719
Epoch 49/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3180 - accuracy: 0.8633 - val_loss: 0.3588 - val_accuracy: 0.8687
Epoch 50/50
320/320 [==============================] - 3s 9ms/step - loss: 0.3025 - accuracy: 0.8773 - val_loss: 0.2891 - val_accuracy: 0.8875

model_evaluation_history = model.evaluate(features_test, labels_test)

13/13 [==============================] - 2s 75ms/step - loss: 0.3062 - accuracy: 0.8750
```

Fig 5. Accuracy of custom model

```
Epoch 1/50
320/320 [==============================] - 4s 10ms/step - loss: 0.6597 - accuracy: 0.6500 - val_loss: 0.6344 - val_accuracy: 0.6250
Epoch 2/50
320/320 [==============================] - 3s 9ms/step - loss: 0.5512 - accuracy: 0.7219 - val_loss: 0.6981 - val_accuracy: 0.5813
Epoch 3/50
320/320 [==============================] - 3s 9ms/step - loss: 0.5447 - accuracy: 0.7281 - val_loss: 0.6934 - val_accuracy: 0.6094
Epoch 4/50
320/320 [==============================] - 3s 9ms/step - loss: 0.5036 - accuracy: 0.7711 - val_loss: 0.4653 - val_accuracy: 0.8000
Epoch 5/50
320/320 [==============================] - 3s 9ms/step - loss: 0.5111 - accuracy: 0.7516 - val_loss: 0.6579 - val_accuracy: 0.6094
Epoch 6/50
320/320 [==============================] - 3s 9ms/step - loss: 0.4898 - accuracy: 0.7641 - val_loss: 0.5534 - val_accuracy: 0.6687
Epoch 7/50
320/320 [==============================] - 3s 9ms/step - loss: 0.4784 - accuracy: 0.7742 - val_loss: 0.4393 - val_accuracy: 0.7406
Epoch 8/50
320/320 [==============================] - 3s 9ms/step - loss: 0.4580 - accuracy: 0.7852 - val_loss: 0.7226 - val_accuracy: 0.5969
Epoch 9/50
320/320 [==============================] - 3s 9ms/step - loss: 0.4627 - accuracy: 0.7930 - val_loss: 0.6284 - val_accuracy: 0.7250
Epoch 10/50
320/320 [==============================] - 3s 9ms/step - loss: 0.4533 - accuracy: 0.7859 - val_loss: 0.7033 - val_accuracy: 0.6031

model_evaluation_history = model.evaluate(features_test, labels_test)

13/13 [==============================] - 1s 23ms/step - loss: 0.5152 - accuracy: 0.7550
```

Fig 6. Accuracy of VGG-16 model

Fig 6 shows that the accuracy of the model is increased from epochs 45 to 50. The Total loss vs Total Validation loss plot of VIFFD trained on VGG-16 has values ranging from 1 to 5. The same plot of VIFFD trained on the custom CNN model has loss values in the range 0.2 to 1.4 which clearly shows that the loss values of the custom CNN model are far lower than the VGG-16 model. The accuracy of the model is dependent on certain factors like

i) Resolution of the image

Higher the resolution of the images, better the results obtained. The reason behind this is when the input image is resized, it would preserve the quality of the image which makes the detection of tampered frames easier. Higher resolution images will have less noise so that it gives the model more data to work with.

ii) Maximum images per class

There are two classes in video forgery detection: either the video is forged or its authentic.The model was trained on few images per class,but increasing the number of images per class may result in better accuracy at the cost of using much more RAM.

iii) Choice of pretrained model used

A part of the model was made of the layers of the pretrained VGG-16 model.This resulted in giving better accuracy for our overall model.

ii)User-Interface

The project uses Pycharm Community Edition 2021.1.1 and Django 3.2.4 as a visual aid for Video forgery detection purposes.Django is a free and open source web framework based on Python that uses a model-template-views architecture. It enables the rapid development of a web application for the detection of video forgeries. Pycharm is an Integrated Development Environment (IDE) that includes a variety of key tools for Python developers that are tightly integrated to produce a pleasant environment for productive Python, web, and data science development. PyCharm has support for Django, which is one of its features. PyCharm can be considered the

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greatest IDE for Django due to its ability to include JavaScript functionality. The Django libraries were imported into PyCharm, and a user-friendly web application was constructed.

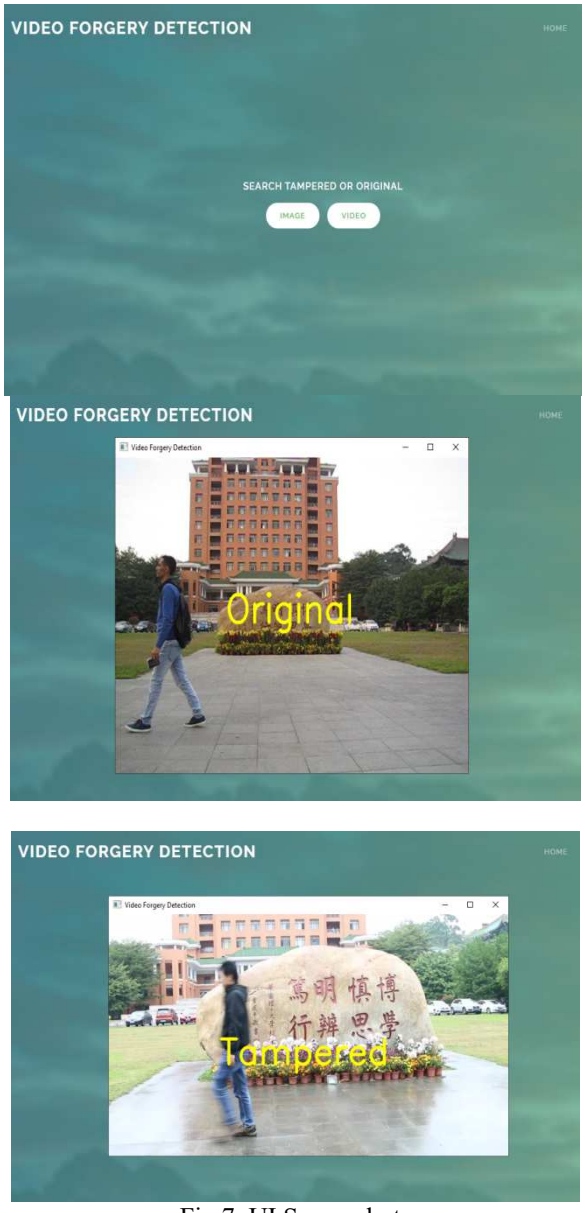

Fig 7. UI Screenshots

V. CONCLUSION

This project is a framework for finding forged frames in a video.This work was able to create a deep learning model which detects forged frames and has managed to achieve a validation accuracy of 87%, which is a significant increase in validation accuracy over the standard VGG-16 which provides 79.5% (11% increase). A user-friendly interface for the novice user was also incorporated for the novice user to be able to use it easily. The proposed model can be further expanded to incorporate features like Localization of forged frames. Also, the achieved accuracy can be further improved by adding more layers and optimising weights. Further the model can be optimised to handle heavy datasets using better hardware infrastructure.

REFERENCES

[1] M. Kobayashi, T. Okabe, and Y. Sato: Detecting video forgeries based on noise characteristics, in Advances Image Video Technology, vol. 5414.Tokyo, Japan: Springer, 2009.

[2] C. Hsu, T.Y. Hung, C.W. Lin, and C.T. Hsu: Video forgery detection using correlation of noise residue, in Proc. IEEE 10th Workshop,Australia, Oct. 2008.

[3] Luca D'Amiano, Davide Cozzolino, Giovanni Poggi: A PatchMatch-Based Dense-Field Algorithm for Video Copy–Move Detection and Localization. IEEE transactions on circuits and systems for video technology, vol. 29, no. 3, March 2019

[4] Lichao Su , Huan Luo, and Shiping Wang: A Novel Forgery Detection Algorithm for Video Foreground Removal, IEEE accessPP(99):1-1,August 2019

[5] Mubashar Sadique, Khurshid Asghar, Usama Ijaz, Muhammad Hussain, Zulfiqar Habib: Spatial Video Forgery Detection and Localization using Texture Analysis of Consecutive Frames, Advances in Electrical and Computer Engineering, August 2019

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